

Ceferino Patino

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Objective

Aspiring computer scientist eager to continue learning and develop practical skills through meaningful industry experience. Seeking opportunities to apply my software development knowledge while contributing to innovative projects.

Education

B.S. in Computer Science <i>University of Nebraska – Lincoln</i>	Lincoln, NE, United States <i>May 2024 – May 2026</i>
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Honors and Awards

Undergraduate Student Researcher Award	University of Nebraska – Lincoln <i>2025</i>
Dean’s List	University of Nebraska – Lincoln <i>Fall 2024 - Spring 2026</i>

Skills

Languages: Python, C++, Rust, JavaScript, Lua, Bash, SQL, Nix, Go, Java
Frameworks & Libraries: PyTorch, OpenCV, Astro, React, Express.js
Tools & Platforms: Docker, K8, Terraform, Git, Nix/NixOS, Maturin, MPI, HDF5, AWS, Django, PostgreSQL, MySQL, sqlite
Concepts: Machine Learning, RL/MARL
Software Engineering: SOLID, Agile, SCRUM, CI/CD, DevOps, Parallel Programming
Soft Skills: Team collaboration, Technical communication, Self-directed learning, Mentorship

Certifications

AWS Certified Cloud Practitioner	Amazon Web Services <i>October 2023 – May 2026</i>
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Experience

Mutual of Omaha <i>Application Developer ← I/S Intern</i>	Remote <i>August 2026 – October 2026</i>
- Developed and maintained enterprise applications using Java and Groovy, building RESTful APIs and backend services to support business operations - Built responsive front-end interfaces with Angular, JavaScript, and TypeScript, improving user experience for internal stakeholders - Designed and optimized complex queries in SQL and Oracle SQL to retrieve and manipulate data across multiple database systems - Collaborated with cross-functional teams to design and implement system enhancements that improved operational efficiency - Maintained and integrated existing applications while ensuring compliance with company Information Services policies and security standards - Implemented appropriate security and privacy measures to protect sensitive company and customer data	
OASYS Lab – University of Nebraska – Lincoln <i>Undergraduate Researcher</i>	Lincoln, NE, United States <i>May 2024 – August 2026</i>
- Contributed to advanced MARL research in a graduate research group, developing a benchmark suite used by 20+ researchers internationally. - Enabled dynamic agent/task/frame adaptation for open-system evaluation; used by 20+ international researchers. - Developed free-range-rust, a Rust+CUDA backend to accelerate free-range-zoo, doubling environment performance. - Integrated Python bindings with Maturin to expose custom CUDA logic for seamless use in RL workflows. - Co-led MOASEI competition at AAMAS 2025 with 10+ teams, benchmarking MARL agents in open-agent/task settings	
Raman Lab – Washington University <i>Software Developer</i>	Remote <i>June 2024 – May 2026</i>
- Developed real-time imaging software for a 3D 2-photon lightsheet microscope processing 20 GB/s of image data. - Designed parallel MPI/HDF5 data acquisition and OpenCV pipelines for volumetric dataset processing.	
CSCE361 Software Engineering – University of Nebraska – Lincoln <i>Undergraduate TA under Dr. Bhuvana Gopal</i>	Lincoln, NE, United States <i>August 2025 – May 2026</i>
- Support students in labs, assignments, and group projects for CSCE361, a course focused on large-scale software development and engineering best practices. - Host technical walkthroughs and office hours to assist with programming and technology tools. - Grade quizzes and exams, and provide feedback on course deliverables. - Facilitate team-based learning and help students understand software engineering concepts, including requirements analysis, design, testing, and maintenance.	
Danforth Plant Science Center <i>Software Developer</i>	St. Louis, MO, United States <i>December 2023 – August 2024</i>
- Developed computer vision software for analysis of underground agricultural data for plant phenotyping. - Worked with developer teams to manage AWS infrastructure for processing high-density aerial crop data.	

Open Source Contributions

nixpkgs

Contributor & Package Maintainer

NixOS Project

May 2025 – Present

- Contribute to nixpkgs with improvements, reviews, and new package additions
- Collaborate with the community to ensure package quality and build reproducibility

nixvim

Maintainer

NixOS Project

May 2025 – Present

- Maintain Nixvim, a modular Neovim compatibility port for Nix
- Add plugin support for popular vim plugins and neovim integrations and improve docs for the community

Turso

Contributor & Package Maintainer

Distributed Edge Database

2025

- Fixed a core bug in Turso, a distributed edge database built on libSQL, improving reliability for production users
- Collaborated with maintainers to review, test, and merge the fix into the main codebase
- Engaged with the open-source community to ensure robust, high-quality contributions

Projects

cpatino.com

Web Development – JavaScript, Astro

Personal

May 2025 – Present

- Built and deployed a personal site with Astro, optimized for minimal bundle size and fast loads

yumevim – “dream vim”

Dev Tooling – Lua, Nix, Neovim

Personal

July 2024 – Present

- Architected a modular, declarative Neovim configuration with Lua and Nix for reproducible setups across devices
- Implemented zero-downtime updates with atomic rollbacks using Nix flakes and Git

3D Volumetric 2-Photon Lightsheet Microscope

Biotechnology / Microscopy – C++, MPI, HDF5, OpenCV

Raman Lab – Washington University

June 2024 – Present

- Developed software for a cutting-edge 3D volumetric 2-photon lightsheet microscope—one of few worldwide
- Built parallel data acquisition pipelines with MPI and HDF5 handling 20 GB/s of image data
- Optimized imaging workflows with OpenCV for real-time processing of volumetric datasets

free-range-zoo

AI / Reinforcement Learning – Python, PyTorch

OASYS Lab – University of Nebraska – Lincoln

April 2024 – Present

- Developed open-environment benchmarks for POSG-based multi-agent RL (wildfire, cybersecurity, rideshare)
- Implemented vectorized training loops for MADDPG, COMA, and GNN-based RL policies
- Enabled dynamic agent/task/frame changes to benchmark adaptability in open systems
- Used by 20+ researchers internationally as a benchmark RL algorithms in multi-agent environments

free-range-rust

AI / Reinforcement Learning – Rust, CUDA, Maturin

OASYS Lab – University of Nebraska – Lincoln

April 2024 – Present

- Implemented custom CUDA kernels in Rust to accelerate dynamic RL environments
- Doubled performance of **free-range-zoo** through low-level vectorized space operations
- Exposed Rust/CUDA modules to Python via Maturin for seamless integration and use by other research groups

yumeami – “dream network”

DevOps – Nix, NixOS

Personal

April 2024 – Present

- Engineered modular NixOS configurations with separate system and home profiles across heterogeneous hardware
- Developed a self-hosted infrastructure integrating services, CI/CD pipelines, and custom tooling for automation
- Implemented declarative provisioning and deployment workflows enabling reproducible, scalable environments

UNL Senior Design Capstone (Mutual of Omaha – Sponsor)

Student Software Engineer

Lincoln, NE, United States

September 2025 – May 2026

- Co-led design and development of a fullstack JSON validation engine for Mutual of Omaha internal workflow automation.
- Built and deployed the solution using React (frontend), Spring Boot (API backend), AWS CDK for infra-as-code, and ECS/Aurora for production cloud delivery.
- Collaborated with Mutual of Omaha engineers to deliver requirements, review code, and build CI/CD pipelines for cloud deployment.

Fonio Seed Computer Vision

Computer Vision – Roboflow, OpenCV

Donald Danforth Plant Science Center

April 2024 – August 2024

- Compiled and labeled a dataset of over 8,000 fonio seed images for robust model training
- Implemented a neural network achieving 99% accuracy in detecting overlapping seeds and providing precise counts

SubTerra.ai

Computer Vision / Agricultural AI – PyTorch, OpenCV, Django, Docker

Donald Danforth Plant Science Center

December 2023 – August 2024

- Developed a Django interface to run segmentation and extrapolate root volume
- Built root segmentation models using PyTorch for high-resolution root image data
- Processed segmentation masks with OpenCV to compute volumetric estimates achieving 97% accuracy

FieldDock

Computer Vision / Agricultural AI – OpenCV, Docker, AWS

Donald Danforth Plant Science Center

December 2023 – May 2024

- Deployed multiple components of a high-throughput drone data collection system using Docker and AWS
- Developed flight software for drone vision and landing leveraging OpenCV

qOverflow

Black Data Processing Associates (BDPA)

Web Development – JavaScript, MongoDB, React, Express.js, Node.js

June 2022 – August 2022

- Won first place in 2022 BDPA hackathon developing a Stack Overflow-inspired Q&A platform with custom analytics
- Containerized the app using Docker and deployed on Kubernetes; documented APIs with SwaggerHub

hypixel-helper

Personal

Web Development – JavaScript, React, Redis

April 2022 – May 2022

- Built live tracking and forecasting of Hypixel Skyblock marketplace data using SARIMA

Publications

Inaugural MOASEI Competition at AAMAS’2025: A Technical Report

arXiv:2507.05469

Patino C., Billings T., Abadi A., Redder D., Eck A., Doshi P., Soh L.

2025